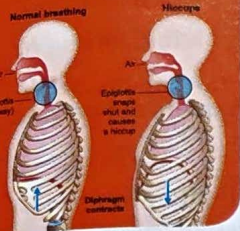


CHAPTER

12



Hikka

Learning Objectives

- ⇒ To learn the concept of Hikka
- ⇒ To learn the Nidana Panchaka of Hikka mentioned in Ayurveda textbooks
- ⇒ To frame out the probable Samprapti Ghataka based on the progression of the disease
- ⇒ To describe the cardinal features of individual type of Hikka

Introduction

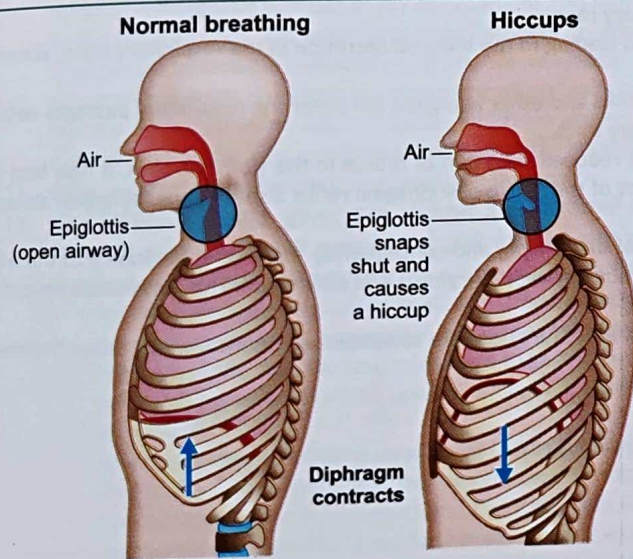


Figure 26 : Image of healthy individual and patient of Hikka

Hikka (see Figure 26) is the spectrum of diseases where Vata Kapha Dosha vitiation is observed in the pathogenesis of the disease. This is characterised by production of specific type of sound 'Hi' frequently as if the organs of abdomen will come out of the mouth. This is afflicting Rasa Dhatu and involves Pranavaha and Rasavaha Srotas.

As Pranavaha Srotas is afflicted in the patient, one can observe the involvement of Hridaya and Mahasrotas in due course of time, because Srotomula affliction can be commonest event that can occur in case of progression of the disease. Hridaya and Mahasrotas may afflict during the pathogenesis of the disease. It is also termed as Hidhma. As per current understanding, hiccup is a phenomenon which occurs as a result of hyperactivity of diaphragm, one of the primary muscle of respiration.

कफवातात्मकावेतौ पित्तस्थान समुद्भवौ।

हृदयस्य रसादीनां धातूनां चोपशोषणौ॥४॥

Ca.Ci. - 17/8

Hikka is considered to be Amashaya Samuttha Vikara, manifested due to the vitiation of Vata and Kapha Dosha, finally resulting in the dryness of Rasa etc Dhatu.

मुहुर्मुहुर्वायुरुदेति सस्वनो यकृद्दिहान्त्राणि मुखादिवाक्षिपन्।

स घोषवानाशु हिनस्त्यसून् यतस्ततस्तु हिक्केत्यभिधीयते बुधैः॥३॥

Ma.Ni. - 12/3

The person will have specific sound 'Hi' frequently through the mouth by vitiated Vata as if Yakrit, Pliha and Antra etc organs of the abdomen come out of the mouth. This is considered to be life threatening illness. As the sound 'Hi' is produced, this disease is named as Hikka.

Hikka is group of the disease where we can observe repeated involuntary spasms of diaphragm followed by sudden closure of the glottis,

which checks the inflow of the air and causes the characteristic sound. Normally, the opening of the glottis synchronises with the contraction of the diaphragm, and consequently there is no hindrance to the free entry of air. Oesophageal disease such as gastro oesophageal reflux disease; abdominal diseases such as bowel diseases, gall bladder disease, hepatitis, pancreatitis, pregnancy; thoracic diseases such as diaphragmatic pleurisy, pericarditis, pneumonia, thoracic surgery and other causes such as alcoholism, uremia and posterior fossa tumours of the brain are considered as the intractable causative factors of the hiccup.

Nidana of Hikka

रजसा धूमवाताभ्यां शीतस्थानाम्बुसेवनात्।
 व्यायामानाम्यधर्माध्वरूक्षान्न विषमाशनात्॥ 1 ॥
 आमप्रदोषादानाहाद्रीक्ष्यादत्यपतर्पणात्।
 दीर्बल्यान्मर्षणो घातद्वन्द्वाच्छुद्ध्यतियोगतः॥ 2 ॥
 अतीसारज्वरच्छर्दि प्रतिश्यायक्षतक्षयात्।
 रक्तपित्तादुदावर्ताद्विसूच्यलसकादपि॥ 3 ॥
 पाण्डुरोगाद्विषाञ्चैव प्रवर्तते गदाविमौ।
 निष्वावमाषपिण्याकतिलतैलनिषेवणात्॥ 4 ॥
 पिष्टशालुकविष्टम्भिविदाहिगुरुभोजनात्।
 जलजानूपपिशितदध्यामक्षीरसेवनात्॥ 5 ॥
 अभिष्यन्द्युपचाराच्च श्लेष्मलानां च सेवनात्।
 कण्ठोरसः प्रतीघाताद्विबन्धैश्च पृथग्विधैः॥ 6 ॥
 विदाहिगुरुविष्टम्भिरूक्षामिष्यन्दिभोजनैः।
 शीतपानासनस्थानरजोधूमानिलानलैः॥ 3 ॥
 व्यायाम कर्मभाराध्ववेगाघातापतर्पणैः।
 आमदोषाभिघातस्त्रीक्षयरोगप्रपीडनैः॥ 4 ॥

विषमाशनाध्यनशनैस्तथा संशमनैरपि।
 हिक्का श्वासश्च कासश्च नृणां समुपजायते॥ 5 ॥
 विदाहिगुरुविष्टम्भिरूक्षामिष्यन्दिभोजनैः।
 शीतपानाशनस्थानरजोधूमातपानिलैः॥ 1 ॥
 व्यायामकर्मभाराध्ववेगाघातापतर्पणैः।

Su.Ut. - 50/3-5

Ma.Ni. - 12/1-2

- ⊗ Excess exposure to dust
- ⊗ Excess exposure to **smoke**²
- ⊗ Excess exposure to cold air
- ⊗ Residing in very cold environment for most of the occasions
- ⊗ Excess intake of **ice cold water**³
- ⊗ **Excess exercise**⁴
- ⊗ Excess indulgence in **sexual intercourse**⁵
- ⊗ Excess intake of **dry food substances**⁶
- ⊗ Frequent consumption of food substances at improper time and in inappropriate quantity
- ⊗ When an individual is suffering *Amapradosha* (*acute*

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- 2 Excessive exposure to these causes are responsible for the manifestation of pneumonia. As mentioned above, pneumonia is considered as one of the intractable cause of hiccup. Exposure to dust and smoke may lead to inflammation of upper and lower respiratory passages. If the size of the particle is $< 6 \mu$, then such particle will be entering the alveoli and suspend in the air and even gets attached to the wall of the alveolus, initiating the inflammatory process.
- 3 When an individual is exposed to cold atmosphere, several physiological changes may occur in the body.
 1. Contraction of mucosa of the respiratory system, skin, and alimentary canal.
 2. This may initiate the activation of sympathetic nervous system, thus leading to the reduced secretion in the respiratory tract, alimentary canal and skin.
 3. Due to the reduced secretion in the above said parts, the dust, smoke and other allergens will enter the respiratory passages easily, thereby leads to the inflammatory changes to occur in the respiratory tract.
 4. Due to the entry of ice cold water, in the alimentary canal, there is reduced secretion of mucus in the stomach, thus, it may lead to alimentary canal dysfunction. In the above passage, it is said that any of the gastro oesophageal reflux disease or oesophageal disease is considered to be the intractable cause of hiccup.
- 4 Normally, the opening of the glottis synchronises with the contraction of diaphragm, and consequently there is no hindrance to the free entry of the air. In case of excess exercise, there may be asynchronous action between the glottis and diaphragm. This may initiate episodes of hiccups.
 When the person is indulging in excess sexual intercourse and if his respiratory functions are compromised, asynchronous action between glottis and diaphragm may occur which initiate the episodes of hiccups.
- 5 In males, the stages of male sexual response cycle are as follow:
 1. Presence of libido or sexual drive
 2. Erection
 3. Emission and ejaculation
 4. Male orgasm
 5. State of resolution
 In females, the stages of female sexual response cycle are as follow:
 1. Excitement phase: Presence of libido including psychic and local sexual stimulation
 2. Plateau phase: Female erection of clitoris and lubrication of external genitals and vagina
 3. Orgasm phase: Female orgasm (extreme satisfaction)
 4. Resolution phase
 It is being mentioned that the best cure for intractable hiccups is sexual intercourse as per Fesmire on his study on rectal massage which was published in Scientist in 2006. It is being responsible for the appearance of orgasm. This orgasm is said to be stimulating parasympathetic nervous system and this stimulation inhibits the overactivity of diaphragm, a primary muscle of respiration. In case of indulgence in sexual intercourse, the appearance of orgasm may diminish in both sexual partners. In the males, ejaculation is considered to be the stage in male orgasm and in females, the orgasm may occur in two or three or more occasions during or after sexual intercourse. Ejaculation of semen into the vagina may not be the reason for female orgasm. When the couple indulge in excess sexual intercourse, that will be leading to the activation of sympathetic nervous system which in turn leads to the reduced secretion of alimentary secretions, and stimulates the overactivity of diaphragm. Thus, the person may develop hiccup. course of time, this is responsible for the reduced secretion of the digestive juices and this may induce the alimentary canal disturbances, which may be considered as one of the intractable cause of hiccups.
- 6 When an individual consumes excess dry substances such as bread etc, that may lead to the activation of sympathetic nervous system in due

affliction of alimentary canal⁷

- ❖ When an individual is suffering from distension of abdomen⁸
- ❖ When a person is exposed to dry environment⁹
- ❖ When an individual is undergoing the treatment measures leading to depletion of body tissues¹⁰ i.e. Langhana therapy
- ❖ When an individual is having severe debility
- ❖ When an person is having injury to Marma (vital points)¹¹
- ❖ Dvandva¹²
- ❖ When an individual is undergoing excess Shodhana (purificatory therapy) line of treatment excessively
- ❖ When an individual is suffering from Atisara, Jvara, Chardi, Pratishyaya, Kshata, Kshaya, Raktapitta, Udavarta, Visuchika and Alasaka, Panduroga and Visha¹³
- ❖ If a person consumes Nishpava (beans), blackgram, Pinyak (oil cake), substances prepared from sesamum

oil, substances prepared from flour, *Shaluka* (esculent root of different kinds of lotus), substances which hampers the movement of *Vata Dosha* in the *Koshtha* (alimentary canal), substances which cause *Vidaha* (burning sensation in the chest and thirst)¹⁴,

- ❖ Substances which are heavy to digest
- ❖ If the person consumes the meat of aquatic and marshy animals
- ❖ If the person consumes curds in excess quantity
- ❖ If the person consumes milk without proper boiling
- ❖ If the person consumes more amount of *Abhishyandi Ahara* (substances which cause obstruction in the *Srotas*)
- ❖ If the person consumes more amount of *Kapha* vitiating substances¹⁵
- ❖ Suppression of natural urges¹⁶
- ❖ If the person consumes excess amount of food before the digestion of previously ingested meals¹⁷

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- 7 In this instance, the person may have either acute inflammation of the upper or lower alimentary canal or both or there may be the possibility of acute irritation of the diaphragm. The outcome of the above said mechanisms may be the occurrence of hiccups.
- 8 When an individual is presented with above said clinical presentation, the person may have either small or large intestinal obstruction or there may be a possibility of irritation of the diaphragm due to the overstretching of the diaphragm because of intestinal obstruction. As stated earlier, bowel diseases are considered as intractable cause of hiccups.
- 9 When an individual is exposed to dry environment, this may lead to the excitation of sympathetic nervous system, this may in turn is leading to the reduced secretion in the respiratory airways or alimentary canal or both. It is being said that pneumonia and gastro oesophageal diseases are intractable causes of manifestation of hiccup.
- 10 When an individual is undergoing treatment which leads to excess lightness of the body, that means the person is undergoing emaciation therapy. In this situation, adipose tissues which are accumulated in different parts of the body, that may enter the general circulation, and these will enter the liver, resulting in hepatomegaly. hepatic diseases are considered as one of the intractable cause of hiccups.
- 11 Refer the reference of undergoing Langhana therapy
- 12 लङ्घनातिथोगादितः। द्वन्द्वात् गुरुलघू द्वन्द्वो शीतोष्णो द्वन्द्वो मन्दतीक्ष्णो द्वन्द्वो मृदुकठिनो द्वन्द्वो खरमसृणो द्वन्द्वो विशदपिच्छिलो द्वन्द्वो स्थिरसरो द्वन्द्वो स्थूलसूक्ष्मो द्वन्द्वो स्निग्धरुक्षी द्वन्द्वो सान्द्रद्रवो द्वन्द्वो सुखदुःखे द्वन्द्वो इन्द्रो चेच्छाद्वंषी द्वन्द्वो प्रवृत्तिनिवृत्ती चेति। अतीसारादयो रोगा हिकाश्चासुरोग Gangadhara; Ca.Ci. – 17/10-16
In this, the term Dvandva indicates the continuous consumption of substances having opposite qualities at a particular time frequently.
 - For example, consumption of Shita and Ushna substances. The consumption of Shita substances at the beginning will lead to the dilution of gastric juice, constriction of mucous cells and less secretion of mucus. At this particular time period, if the person consumes hot substances, this may lead to damage to gastric mucosa. This may lead to the irritation of the alimentary canal leading to the excess peristaltic movements in the upward direction, causing vomiting. When the person is having vomiting, it is initiated by contraction of diaphragm.
 - Suppose, if the individual is consuming Ushna substances at the beginning itself, this may result in inflammation of stomach. This will lead to the reverse peristaltic movement, causing vomiting. At this time period, if the person consumes Shita (cold) substances, this may be responsible for constriction of mucous cells of the stomach, reduced gastric juice secretion. As there is inflammation in the stomach, the person may have further episodes of vomiting. This may lead to contraction of diaphragm.
 - Thus consumption of Shita and Ushna substances or Ushna and Shita substances consequently may lead to the initiation and worsening of vomiting. GERD is considered to be one of the intractable cause of vomiting.
 - As per French Index of differential diagnosis textbook, sudden change in the temperature of stomach either consumption of very cold or hot substances induces hiccups.
- 13 Refer the reference of undergoing Langhana therapy.
- 14 These substances will be responsible for the vitiation of Vata and Pitta Dosha in the body.
- 15 The exposure to above said aetiological factors will lead to the vitiation of the Kapha Dosha in the body. When Kapha Dosha is increased in the body, this will lead to the excess respiratory secretion and secretions from alimentary canal. Thus, both are the risk factors for the development of inflammatory changes in the respiratory airways and alimentary canal. As earlier stated, pneumonia and GERD are the intractable causes of hiccups.
- 16 Suppose, let us take the example of Purisha Vega Dharana (suppression of urge of defaecation). When the person is frequently associated with suppression of urge of defaecation, that will lead to the vitiation of Apanavata, which in turn leads to the abnormal functioning of Samana Vata and later there will be affliction of Prana Vata can be seen. If the individual is having the frequent habit of suppression of urge of defaecation, that may lead to excess distension of the abdomen, and this may be the cause of irritation of diaphragm and thus the person may develop hiccups.
- 17 In this situation, the person may have excess vitiation of Kapha Dosha in the body. Refer the causes of vitiation of Kapha Dosha and sub-

Samprapti of Hikka

मारुतः प्राणवाहीनि स्रोतांस्याविश्य कुप्यति।

उरःस्थः कफमुद्ध्य हिक्काश्वासान् करोति सः॥17॥ Ca.Ci. – 17/17

प्राणोदकात्रवाहीनि स्रोतांसि सकफोऽनिलः।

हिक्काः करोति संरुध्य तासां लिङ्गं पृथक् शृणु 21 Ca.Ci. – 17/21

⊗ Vata and Kapha Dosha get vitiated involving Pranavaha, Udakavaha and Annavaha Srotas hampering their function.

⊗ Hikka is manifested due to this abnormal mechanism.

Samprapti Ghataka

⊗ Dosha : Vata Kapha

⊗ Dushya : Rasa, Udaka

⊗ Srotas : Pranavaha, Udakavaha and Annavaha Srotas

⊗ Srotodushti Prakara : Sanga and Vimargagamana

⊗ Agni : Jatharagnimandya

⊗ Ama : Jatharagnimandyaajanya

⊗ Udbhava Sthana : Amashaya

⊗ Sanchara Sthana : Sarva Sharira (as Rasa Dhatu is involved)

⊗ Vyakta Sthana : Vaktra and Uras

⊗ Rogamarga : Abhyantara and Bahya

⊗ Vyadhi Svabhava : Ashukari

⊗ Sadhyasadhyata : Asadhya

Purvarupa

कण्ठोरसोर्गुरुत्वं च वदनस्य कषायता।

हिक्कानां पूर्वरूपाणि कुक्षेरादोष एव च॥19॥

Ca.Ci. – 17/19

मुखं कषायमरतिगौरवं कण्ठवक्षसोः।

पूर्वरूपाणि हिक्कानामादोषो जठरस्य च॥18॥

Su.Ut. – 50/8

⊗ Heaviness of the throat¹⁸

⊗ Heaviness of the chest¹⁹

⊗ Astringent taste of the mouth²⁰

⊗ Distension of the abdomen²¹

stances having Guru quality (heavy to digest) etc.

18 Normally, closure of the glottis synchronises with the contraction of the diaphragm, and consequently there is no hindrance to the free entry of air. When the individual is having asynchronous function of glottis and diaphragm, that is responsible for the manifestation of hiccup. This prodromal symptom indicates the asynchronous closure of the glottis. This indicates the undue awareness of dysfunction of glottis and the person may feel heaviness of the throat.

19 Normally, closure of the glottis synchronises with the contraction of the diaphragm, and consequently there is no hindrance to the free entry of air. Heaviness of the chest implies the asynchronous closure of the diaphragm in relation to closure of the glottis.

20 The presence of astringent taste in the mouth indicates the dominance of Vata Dosha at that particular point of time. Vitiation of Vata Dosha may lead to abnormal peristaltic movement of the oesophagus and stomach; autonomic dysfunction which may lead to abnormal motility and reduced secretion. This may lead to asynchronous closure of the glottis and contraction of diaphragm, thus may lead to the manifestation of the hiccup.

21 As far as the functions of Vata in relation to alimentary canal are concerned, the movement of Vata starts from proximal to distal side i.e. from oral cavity to anal region. When a person is presented with distension of the abdomen, Apana Vata affliction occurs at the beginning. When Prana Vata affliction is noted in the individual, that may lead to the abnormal irritation of Prana Vata in the pa-

22 The presence of continuous Hik sound indicates structural abnormalities such as malignant lymphadenopathy from oesophageal cancer;

Types of Hikka

5 types as per Charaka

1. Maha Hikka
2. Gambhira Hikka
3. Vyapeta Hikka
4. Kshudra Hikka
5. Annaja Hikka

5 types as per Sushruta and Madhavakara

1. Annaja Hikka
2. Yamala Hikka
3. Kshudra Hikka
4. Gambhira Hikka
5. Maha Hikka

Maha Hikka

क्षीण मांसबलप्राणतेजसः सकफोऽनिलः।

गृहीत्वा सहसा कण्ठमुच्चैर्घोषवतीं भृशम्॥22॥

करोति सततं हिक्कामेकद्वित्रिगुणां तथा।

प्राणः स्रोतांसि मर्माणि संरुध्योष्माणमेव च॥23॥

संज्ञां मुष्णाति गात्राणां स्तम्भं संजनयत्यपि।

मार्गं चैवान्नपानानां रुणच्युपहतस्मृतेः॥24॥

साश्रुविलुतनेत्रस्य स्तब्धशङ्खच्युतध्रुवः॥

सतजल्पप्रलापस्य निर्वृतिं नाधिगच्छतः॥25॥

महामूला महावेगा महाशब्दा महाबला॥

महाहिकेति सा नृणां सद्यः प्राणहरा मता॥26॥

Ca.Ci. – 17/22-26

मर्माण्यापीडयन्तीव सततं या प्रवर्तते।

देहमायस्य वेगेन घोषयत्यतितृष्यतः॥14॥

महाहिकेति सा ज्ञेया सर्वगात्रप्रकम्पिनी॥15॥

Su.Ut. – 50/14,15; Ma.Ni. – 12/9

⊗ Vata and Kapha Dosha get vitiated in an individual who is physically weak, profound muscular atrophy, poor lustre and afflict throat.

⊗ Production of continuous 'Hik' sound²² twice or

thrice at a time thereby resulting in a life threatening situation.

- ⊗ Altered sensation of the body
- ⊗ Stiffness of the body
- ⊗ Watering from the eyes
- ⊗ Stiffness of the temples and eyebrows
- ⊗ Incomprehensible speech
- ⊗ This disease is deep seated, severe episode of the illness and produces loud sound and is considered to be having more *Rogabala* (strength of the disease is more)

Gambhira Hikka²³

हिकते यः प्रवृद्धस्तु कृशो दीनमना नरः।

जजरिणोरसा कृच्छं गम्भीरमनुनादयन्॥27॥

संजुम्भन् संक्षिपंश्चैव तथाऽङ्गानि प्रसारयन्।

पार्श्वे चोभे समायम्य कूजन् स्तम्भरुगर्दितः॥28॥

नाभेः पकाशयाद्वाऽपि हिक्का चास्योपजायते।

क्षोभयन्ती भृशं देहं नामयन्तीव ताम्यतः॥29॥

रुणच्युच्छ्वासमार्गं तु प्रनष्टबलचेतसः।

गम्भीरा नाम सा तस्य हिक्का प्राणान्तिकी मता॥30॥

Ca.Ci. – 17/27-30

नाभिप्रवृत्ता या हिक्का घोरा गम्भीरनादिनी॥12॥

शुष्ककण्ठजिहास्यश्वासपावरुजाकरी।

अनेकोपद्रव्ययुता गम्भीरा नाम सा स्मृता॥13॥

Su.Ut. – 50/12,13; Ma.Ni. – 12/9

- ⊗ The person is very emaciated and having seriously affliction of chest of the individual.
- ⊗ There will be severe fatigue, emaciation, bodyache
- ⊗ Stiffness of both sides of the chest with crying type of sound
- ⊗ 'Hik' sound is produced from below the umbilicus or from *Pakvashaya*
- ⊗ The person will have bending of the body and will have darkness in front of the eyes
- ⊗ Obstruction of the respiratory passages and the patient will have profound loss of strength and enthusiasm.
- ⊗ This is a life threatening illness.

Vyapeta Hikka²⁴

व्यपेता जायते हिक्का याऽन्नपाने चतुर्विधे।

आहारपरिणामान्ते भूयश्च लभते बलम्॥31॥

प्रलापवम्यतीसारतृष्णार्तस्य विचेतसः।

जृम्भणो विष्टुताक्षस्य शुष्कास्यस्य विनामिनः॥32॥

पर्याध्मातस्य हिक्का या जनुमूलादसन्तता।

सा व्यपेतेति विज्ञेया हिक्का प्राणोपरोधिनी॥33॥

Ca.Ci. – 17/31-33

- ⊗ This type of *Hikka* manifested due to the consumption of 4 types of food.
- ⊗ This is aggravated after the complete digestion of food.
- ⊗ This is associated with irrelevant speech, vomiting, diarrhoea, thirst and altered state of consciousness, yawning, eyes filled with tears, dryness of mouth, forward bending of the body, distension of the abdomen
- ⊗ This is considered to be life threatening illness.

Kshudra Hikka²⁵

क्षुद्रवातो यदा कोष्ठाध्यायामपरिघटितः।

कण्ठे प्रपद्यते हिक्कां तदा क्षुद्रां करोति सः॥34॥

अतिदुःखा न सा चोरः शिरोमर्मप्रबाधिनी।

न चोच्छ्वासात्रपानानां मार्गमावृत्य तिष्ठति॥35॥

वृद्धिमायस्यतो याति भुक्तमात्रे च मार्दवम्।

यतः प्रवर्तते पूर्वं तत एव निवर्तते॥36॥

हृदयं क्लोम कण्ठं च तालुकं च समाश्रिता।

साहिति नृणां साध्या प्रकीर्तिता॥37॥

Ca.Ci. – 17/34-37

विकृष्टकालैर्या वेगैर्मन्दैः समभिवर्तते॥1॥

क्षुद्रिका नाम सा हिक्का जनुमूलात् प्रधाविता।

Su.Ut. – 50/11,12; Ma.Ni. – 12/8

- ⊗ In this type of *Hikka*, mild vitiation of *Vata Dosha* is noted due to physical exertion and vitiation of *Vata* is noted in *Koshtha*.
- ⊗ This is responsible for the affliction of throat and there will be mild form of 'Hik' sound.
- ⊗ The illness is mild and it will not afflict the chest, head and other *Marmas*.

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K

lymphoma; pneumonia of lower lobe of lungs; empyema due to diaphragmatic pleurisy; diaphragmatic hernia; subphrenic abscess; lesions of medulla oblongata; intracranial hemorrhage or brain abscess; uremia or hysteria. Here, it indicates the presence of the hiccup throughout the lifetime of the patient.

23 In this clinical condition, the person may have major disease afflicting the large intestine where chronic constipation may be associated such as carcinoma of sigmoid colon. It does not mean that only the disease of the sigmoid colon may be understood. Because we can consider the *Pakvashaya* as the whole large intestine commencing from caecum to rectum. Thus, from any of the affliction in the large intestine can be considered in this situation.

24 It is being said that intestinal obstruction or surgery related to large intestinal obstruction may be associated with hiccups. In this situation, there may be the possibility of obstruction at splenic or hepatic flexure of large intestine which has close association with diaphragm.

25 In this situation, rapid eating of food or dry food with little amount of water may be considered as the reason for the manifestation of hiccups. Here, the peculiarity about the pathogenesis is restricted to alimentary canal.

- This will not obstruct the passages of air and food.
- This disease will subside after the consumption of food.
- This is a self limiting disease.
- This involves *Hridaya* (heart), *Kloma* (site of thirst), *Kantha* (throat) and *Talu* (palate) and is mild in nature and is curable disease.

Annaja Hikka²⁶

सहसाऽत्यभ्यवहृतैः पानान्नैः पीडितोऽनिलः।

ऊर्ध्वं प्रपद्यते कोष्ठान्मद्यैर्वाऽतिमदप्रदेः॥३८॥

तथाऽतिरोषभाष्याध्वहास्यभारातिवर्तनैः।

वायुः कोष्ठगतो धावन् पानभोज्यप्रपीडितः॥३९॥

उरः स्रोतः समाविश्य कुर्याद्विक्रान्तं ततोऽन्नजाम्।

तथा शनैरसंबन्धं क्षुवंश्चापि स हिक्कते॥४०॥

न मर्मबाधाजननी नेन्द्रियाणां प्रबाधिनी।

हिक्का पीते तथा भुक्ते शमं याति च साऽन्नजा 41 Ca.Ci. – 17/38-41

त्वरमाणस्य चाहारं भुञ्जानस्याथवा घनम्।

वायुरन्नैरवस्तीर्णः कटुकैरर्दितो भृशम्॥९॥

हिक्कत्यूर्ध्वगो भूत्वा तां विद्यादन्नजां भिषक्। Su.Ut. – 50/9,10

पानान्नैरतिसंयुक्तैः सहसा पीडितोऽनिलः।

हिक्कयत्यूर्ध्वगो भूत्वा तां विद्यादन्नजां भिषक्॥६॥ Ma.Ni. – 12/6

- This type of *Hikka* is manifested due to the vitiation of *Vata* which occurs due to the improper food habits, excess intake of alcoholic beverages, excess anger, excess speech, excess walk, laughing and lifting heavy weights.
- This will lead to the vitiation of *Vata* in *Koshtha*.
- This will lead to the forceful movement of *Vata* upwards leading to the production of '*Hik*' sound. This will not afflict the *Marma* (vital parts) and *Indriya* (sense organs).

- By drinking water and eating food, this illness will subside.
- This is a mild and not a life threatening disease.

Yamala Hikka²⁷

चिरेण यमलैर्वागैर्या हिक्का संप्रवर्तते॥१०॥

कम्पयन्ती शिरोग्रीवं यमलां तां विनिर्दिशेत्।

Su.Ut. – 50/10,11; Ma.Ni. – 12/7

- This type of *Hikka* can be seen as production of sound '*Hik*' twice at one episode.
- This will be responsible for involuntary movement of head and neck.

Asadhya Lakshana of Hikka²⁸

आयम्यते हिक्कतो यस्य देहो दृष्टिश्चोर्ध्वं नाभ्यते यस्य नित्यम्।

क्षौति यश्चातिमात्रं तौ द्वौ चान्त्यौ वर्जयेद्विक्रमानौ॥

अतिसंचितदोषस्य भक्तच्छेदकृशस्य च।

व्याधिभिः क्षीणदेहस्य वृद्धस्यातिव्यवायिनः॥१२॥

आसां या सा समुत्पन्ना हिक्का हन्त्याशु जीवितम्।

यमिका च प्रलापार्तिमोहतृष्णासमन्विता॥१३॥

अक्षीणश्चाप्यदीनश्च स्थिरधात्विन्द्रियश्च यः।

तस्य साधयितुं शक्या यमिका हन्त्यतोऽन्यथा॥१४॥

Ma.Ni. – 12/11-14

- If the person has bending of his or her body with upward gaze frequently²⁹
- If the person is having depletion of body tissues³⁰
- If the person is having aversion to food substances³¹
- If the person is having hiccups frequently³²
- If the person is having excess vitiation of *Dosha*
- If the person is suffering from chronic debilitating illness
- If the person is elderly individual³³

- G K**
- 26 In this situation, the person may have emotional factors too along with abnormal food patterns as well as extreme physical activities. This may stimulate abnormal and increased stimulation of diaphragm or phrenic nerve.
 - 27 This is considered to be milder form of hiccups.
 - 28 In any type of *Hikka*, if there is association of *Vata Dosha* vitiation in particular, then such type of *Hikka* is considered as *Asadhya* (incurable) in nature.
 - 29 This is the condition where the person will have bending (forward or backward) of the body like a bow associated with hiccups. In this situation, the person will have upward gaze continuously or frequently.
 - 30 When an individual is having excess depletion of the body tissues implies the person is having deficiency of essential nutrients – carbohydrates, fats and proteins and later the person develops the state of metabolic acidosis either ketoacidosis or lactic acidosis. If the person is having episodes of hiccup at this stage, it further worsens the state of metabolic acidosis, because of further rapid nutrient loss.
 - 31 When an individual is having episodes of hiccups associated with aversion to take food substances, that may lead to metabolic acidosis either ketoacidosis or lactic acidosis. In this instance, the appearance of hiccups may further deteriorate the condition of the patient.
 - 32 The prognosis of hiccups in this situation depends upon the nature of the cause of the disease manifestation. First of all, identify the cause of the disease. As mentioned earlier in the chapter, Oesophageal disease such as gastro oesophageal reflux disease; abdominal diseases such as pneumonia, thoracic surgery and other causes such as alcoholism, uremia and posterior fossa tumours of the brain are considered as the intractable causative factors of the hiccup. Identify the actual cause and give the appropriate treatment.
 - 33 When the person is having hiccups associated with above said situations, the person will have excessive nutrition loss from the body, which may induce the state of metabolic acidosis either ketoacidosis or lactic acidosis. This state may further deteriorate the condition of the patient.

- ⊗ If the person is indulged in **excess sexual intercourse**³⁴
- ⊗ If the person is having **Yamala Hikka** associated with **irrelevant speech**³⁵ or **pain**³⁶ or **confusion**³⁷ or **thirst**³⁸
- ⊗ Above situations are considered to **Asadhya** (incurable)
- ⊗ If the person is suffering from **Yamala Hikka** and he is not emaciated or not having severe symptoms and he is having properly functioning **Dhatu** and **Indriya**, then such **Hikka** is considered to be **Sadhya** (curable)³⁹

Key summary of the chapter

- ⊗ Hikka is the spectrum of the diseases where we are going to observe the Vata and Kapha Dosha vitiation during the pathogenesis. This is characterised by production of specific type of sound 'Hik' frequently as if the organs of the abdomen will come out of the mouth. The verse is mentioned below.

मुहुर्मुहुर्वायुरुदेति सखनो यकृत्हिन्त्राणि मुखादिवाक्षिपन्। स घोषवानाशु हिनस्त्यसून् यतस्ततस्तु हिक्केत्यभिधीयते बुधैः॥३॥

Ma.Ni. – 12/3

- ⊗ Hikka afflicts Rasa Dhatu and involves Pranavaha and Rasavaha Srotas with Sanga and Vimargagamana as the type of Srotodushti observed during the Samprapti of the disease. As pranavaha Srotas affliction is seen, there is a possibility of the involvement of Hridaya and Mahasrotas can be observed sooner or later during the progression of the disease.
- ⊗ As far as Nidana (aetiological factors) of the disease are concerned, abnormal Vata and Kapha Dosha vitiation observed in the Pranavaha Srotas. Rasavaha Srotas and Annavaha Srotas affliction may be observed in due course of time.
- ⊗ As far as Purvarupa of Hikka are concerned, heaviness of the throat and chest; astringent taste of the mouth and distension of the patients.
- ⊗ Sankhya Samprapti of Hikka are of 5 types. As per Charaka, 5 types of Hikka are Maha Hikka, Gambhira Hikka, Vyapeta Hikka, Kshudra Hikka and Annaja Hikka. As per Sushruta and Madhavakara, 5 types of Hikka are Annaja Hikka, Yamala Hikka, Kshudra Hikka, Gambhira Hikka and Maha Hikka. Among 5 types, Gambhira Hikka and Maha Hikka are considered as Asadhya (incurable) in nature.
- ⊗ Normally, the opening of the glottis synchronises with the contraction of the diaphragm, and consequently there is no hindrance to the free entry of air. Hiccups are manifested due to asynchronicity with the functions of glottis and diaphragm. Gastro oesophageal disease, abdominal diseases such as hepatitis, gall bladder disease, pancreatitis; thoracic diseases such as diaphragmatic pleurisy, pericarditis, pneumonia and abdominal or thoracic surgery; Posterior fossa tumours of the brain, alcoholism or uremia. The above said causes are considered as intractable causes of manifestation of hiccups. The term intractable refers to untraceable causes of hiccups.

Self-Assessment: Questions for Review

Short essays

- ⊗ Explain Samprapti and Samprapti Ghataka of Hikka
- ⊗ Explain the Asadhya Lakshana of Hikka
- ⊗ Describe Vyapeta Hikka
- ⊗ Explain Sadhyasadhya of Hikka

MCQs

1. Which of the following is the type of Hikka?

- a. Yamala; b. Vataja;

- c. Pittaja; d. Kaphaja

Answer: (a)

2. Dosha dominance observed in the manifestation of Hikka are

- a. Vata Kapha; b. Pitta Kapha;
c. Vata Rajas; d. Kapha Tama

Answer: (a)

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K

- 34 Normally, the opening of the glottis synchronises with the contraction of the diaphragm, and consequently there is no hindrance to the free entry of air. When the person indulges sexual intercourse in the episode of the hiccup, it indicates the person is performing intercourse in the state of asynchronous action of glottis and diaphragm. This may further worsen the episodes of hiccups as this may alter respiratory rate, rhythm and depth of the breathing along with emotional lability during the sexual intercourse.
- 35 Posterior fossa tumours of the brain, alcoholism or uremia are considered as intractable causes of hiccup. Above said causes may induce irrelevant speech.
- 36 Gastro oesophageal disease, abdominal diseases such as hepatitis, gall bladder disease, pancreatitis; thoracic diseases such as diaphragmatic pleurisy, pericarditis, pneumonia and abdominal or thoracic surgery are associated with pain. The above said causes are considered as intractable causes of manifestation of hiccups.
- 37 Refer the context of irrelevant speech.
- 38 If the person will have severe form of thirst, the person will have syncopal episodes and above said causes in the context of irrelevant speech may worsen the situation of the patient of hiccup.
- 39 When the person is not having any form of Dhatu Kshaya, it implies he will not have sequential depletion of carbohydrates, fats and proteins and electrolytes.

3. Yamala Hikka is mentioned by

- Sushruta;
- Madhavakara;
- Both Sushruta and Madhavakara;
- Charaka

Answer: (c)

4. Which of the following term is associated with Hikka?

- Dvishtarthasamyogaja;
- Nichaya;
- Jara;
- Annaja

Answer: (d)

5. Which of the following term is associated with Hikka?

- Alasa;
- Nirvritti;
- Hidhma;
- Nischaraka

Answer: (c)

6. Vak Stambhana is the Lakshana seen in which of the following?

- Yamala Hikka;
- Chinna Shvasa;
- Maha Hikka;
- Kshudra Shvasa

Answer: (c)

7. Sarvagatra Vikampini Hikka is

- Yamala Hikka;
- Gambhira Hikka;
- Maha Hikka;
- Annaja Hikka

Answer: (c)

8. Gambhira Hikka is

- Sukhasadhya;
- Krichrasadhya;
- Yapya;
- Pratyakhyeya

Answer: (d)

9. Vyapeta Hikka is the contribution of

- Charaka;
- Sushruta;
- Vagbhata;
- Kashyapa

Answer: (a)

10. Hikka is considered as a Vyapat of

- Vamana;
- Virechana;
- Basti;
- Nasya

Answer: (c)

11. Which of the following Hikka is considered as Sadhya (curable)?

- Vyapeta;
- Annaja;
- Gambhira;
- Maha

Answer: (b)

12. The primary muscle involved in the manifestation of hiccup is

- Intercostal muscles;

- Pectoralis major;
- Sternocleidomastoid;
- Diaphragm

Answer: (d)

13. The description of Hikka in Sushrta Samhita is seen in

- Sutra Sthana;
- Nidana Sthana;
- Chikitsa Sthana;
- Uttara Tantra

Answer: (d)

14. Charaka described which of the following disease along with Hikka?

- Kasa;
- Shvasa;
- Rajyakshma;
- Kshata Kshina

Answer: (b)

15. As per Charaka, the site of Hikka is

- Phupphusa;
- Amashaya;
- Pakvashaya;
- Uras

Answer: (b)

Chapter resources

- <https://www.facebook.com/100063775255629/posts/114842453645232/>
- Charaka Samhita, Chikitsa Sthana – 17th chapter – Hikkashvasa Chikitsa Adhyaya
- Sushruta Samhita, Uttara Tantra – 50th chapter – Hikka Pratishedha Adhyaya
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- The Merck Manual of Diagnosis and Therapy by Robert S Porter and Justin L Kaplan, 19th edition, Merck Sharp & Dohme Corp, New Jersey, 2011
- Chamberlain's symptoms and signs in clinical medicine by Colin Ogilvie and Christopher C Evans, 12th edition, Butterworth-Heinemann Publishers, 1997
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